

Let $P(x) = x^7 + a_6x^6 + \cdots + a_0$ be a monic degree-7 polynomial with integer coefficients. The conditions are:

- every coefficient a_i is odd,
- $|a_i| < 2025^2$ for all i ,
- each $|a_i|$ is a root of P .

Set $b_i = |a_i|$ for $i = 0, \dots, 6$ and $b_7 = 1$ (the leading coefficient). All b_i are odd positive integers, $b_i < 2025^2$. Let $M = \max\{b_0, \dots, b_6, 1\}$. M is odd and $M \geq 1$.

Since M is among the b_i , it is a root, so $P(M) = 0$. Then

$$a_0 = -M^7 - a_6M^6 - \cdots - a_1M = -M(M^6 + a_6M^5 + \cdots + a_1),$$

so $M \mid a_0$. Because $|a_0| \leq M$ and a_0 is odd, we must have $|a_0| = M$; hence $a_0 = \pm M$.

The leading coefficient 1 gives $b_7 = 1$, so 1 is a root: $P(1) = 0$. Therefore we can write

$$P(x) = (x-1)Q(x), \quad Q(x) = x^6 + q_5x^5 + \cdots + q_0, \quad q_0 = -a_0 = \mp M.$$

Also M is a root, so $P(x) = (x-M)R(x)$ with R monic of degree 5 and integer coefficients. Combining the two factorizations,

$$P(x) = (x-1)(x-M)S(x),$$

where $S(x) = x^5 + s_4x^4 + s_3x^3 + s_2x^2 + s_1x + s_0$ is monic and has integer coefficients. The constant term of S is $s_0 = a_0/M = \pm 1$.

Expanding the product $(x-1)(x-M)S(x)$ and equating coefficients yields

$$\begin{aligned} a_6 &= s_4 - (M+1), \\ a_5 &= s_3 - (M+1)s_4 + M, \\ a_4 &= s_2 - (M+1)s_3 + Ms_4, \\ a_3 &= s_1 - (M+1)s_2 + Ms_3, \\ a_2 &= s_0 - (M+1)s_1 + Ms_2, \\ a_1 &= -(M+1)s_0 + Ms_1, \\ a_0 &= Ms_0. \end{aligned}$$

Now determine the possible positive integer roots of P . From $P(x) = (x-1)(x-M)S(x)$, any integer root is either 1, M , or a root of S . Because S is monic with constant term ± 1 , the Rational Root Theorem forces any integer root of S to be ± 1 . Hence the only positive integer roots of P are 1 and M . Since each $|a_i|$ is a positive integer root, we must have

$$|a_i| \in \{1, M\} \quad \text{for all } i = 0, \dots, 6.$$

We now treat the two possibilities $M = 1$ and $M > 1$.

0.0.1 Case $M = 1$

Then all $|a_i| = 1$, so each $a_i = \pm 1$. The condition $P(1) = 0$ gives

$$1 + a_6 + a_5 + a_4 + a_3 + a_2 + a_1 + a_0 = 0 \implies a_6 + \cdots + a_0 = -1.$$

Among the seven coefficients $a_6, \dots, a_0$, let k be the number of -1 's. Their sum is $7 - 2k$. Setting $7 - 2k = -1$ yields $k = 4$. Hence exactly four coefficients are -1 and three are $+1$. The number of such polynomials is $\binom{7}{4} = 35$. All satisfy the required conditions (coefficients are odd, $|a_i| = 1 < 2025^2$, and 1 is a root). So this case contributes 35 good polynomials.

0.0.2 Case $M > 1$

Now M is an odd integer ≥ 3 . From (2) we have $|a_i| \in \{1, M\}$ for all i , and $a_0 = \pm M$.

We solve the system (1) under the condition $a_i \in \{\pm 1, \pm M\}$ and $M > 1$.

5.1 Determining s_1 From $a_1 = -(M+1)s_0 + Ms_1$ and $a_1 \in \{\pm 1, \pm M\}$, test possibilities. If $s_0 = 1$: $a_1 = -(M+1) + Ms_1$. For $a_1 = M$ we get $s_1 = 2 + \frac{1}{M}$ (not integer); for $a_1 = -M$ we get $s_1 = \frac{1}{M}$; for $a_1 = 1$ we get $s_1 = 1 + \frac{2}{M}$; for $a_1 = -1$ we get $s_1 = 1$. Thus only $a_1 = -1, s_1 = 1$ works. If $s_0 = -1$: $a_1 = (M+1) + Ms_1$. For $a_1 = 1$ we get $s_1 = -1$; the other possibilities give non-integer s_1 . Hence

$$(s_0, a_1, s_1) = \begin{cases} (1, -1, 1), \\ (-1, 1, -1). \end{cases}$$

5.2 Determining s_2 From $a_2 = s_0 - (M+1)s_1 + Ms_2$ and $a_2 \in \{\pm 1, \pm M\}$.

- For $(s_0, s_1) = (1, 1)$: $a_2 = 1 - (M+1) + Ms_2 = -M + Ms_2 = M(s_2 - 1)$. Because $M > 1$, $M(s_2 - 1) = \pm 1$ would force $M = 1$; hence $s_2 - 1 = \pm 1$, so $s_2 = 2$ or 0 . Then $a_2 = M$ or $a_2 = -M$ respectively.
- For $(s_0, s_1) = (-1, -1)$: $a_2 = -1 - (M+1)(-1) + Ms_2 = -1 + M + 1 + Ms_2 = M + Ms_2 = M(s_2 + 1)$. Similarly $M(s_2 + 1) = \pm M$ gives $s_2 + 1 = \pm 1$, so $s_2 = 0$ or $s_2 = -2$. Then $a_2 = M$ or $a_2 = -M$ respectively.

Thus the admissible pairs are

(s_0, s_1)	(a_2, s_2)
$(1, 1)$	$(M, 2)$ or $(-M, 0)$
$(-1, -1)$	$(M, 0)$ or $(-M, -2)$

5.3 Determining s_3 From $a_3 = s_1 - (M+1)s_2 + Ms_3$ and $a_3 \in \{\pm 1, \pm M\}$.

- **Line (i):** $(s_0, s_1) = (1, 1), s_2 = 2$. Then $a_3 = 1 - 2(M+1) + Ms_3 = -2M - 1 + Ms_3$. $Ms_3 = a_3 + 2M + 1$. For $a_3 = -1$ we get $Ms_3 = 2M$, so $s_3 = 2$ (integer). The other possibilities give non-integer s_3 . Hence $a_3 = -1, s_3 = 2$.
- **Line (ii):** $(s_0, s_1) = (1, 1), s_2 = 0$. Then $a_3 = 1 - 0 + Ms_3 = 1 + Ms_3$. $Ms_3 = a_3 - 1$. For $a_3 = 1$ we get $s_3 = 0$; other choices give non-integer s_3 . Hence $a_3 = 1, s_3 = 0$.
- **Line (iii):** $(s_0, s_1) = (-1, -1), s_2 = 0$. Then $a_3 = -1 - 0 + Ms_3 = -1 + Ms_3$. $Ms_3 = a_3 + 1$. For $a_3 = -1$ we get $s_3 = 0$; other choices give non-integer s_3 . Hence $a_3 = -1, s_3 = 0$.
- **Line (iv):** $(s_0, s_1) = (-1, -1), s_2 = -2$. Then $a_3 = -1 - (M+1)(-2) + Ms_3 = -1 + 2(M+1) + Ms_3 = 2M + 1 + Ms_3$. $Ms_3 = a_3 - 2M - 1$. For $a_3 = 1$ we get $Ms_3 = -2M$, so $s_3 = -2$; other choices give non-integer s_3 . Hence $a_3 = 1, s_3 = -2$.

Thus we obtain the following admissible combinations:

(s_0, s_1, s_2, s_3)	a_1	a_2	a_3
$(1, 1, 2, 2)$	-1	M	-1
$(1, 1, 0, 0)$	-1	$-M$	1
$(-1, -1, 0, 0)$	1	M	-1
$(-1, -1, -2, -2)$	1	$-M$	1

5.4 Determining s_4 and the remaining coefficients Now use $a_4 = s_2 - (M+1)s_3 + Ms_4$ and $a_6 = s_4 - (M+1)$ with $a_4, a_6 \in \{\pm 1, \pm M\}$.

- **Combination (I):** $(s_0, s_1, s_2, s_3) = (1, 1, 2, 2)$. Then $a_4 = 2 - 2(M+1) + Ms_4 = -2M + Ms_4 = M(s_4 - 2)$. Since $M > 1$, $M(s_4 - 2) = \pm 1$ would force $M = 1$; hence $s_4 - 2 = \pm 1$, so $s_4 = 3$ or 1 .

- If $s_4 = 3$, then $a_6 = 3 - (M + 1) = 2 - M$. For $M > 1$ this is in $\{\pm 1, \pm M\}$ only when $M = 3$ (giving $a_6 = -1$), but then $a_5 = s_3 - (M + 1)s_4 + M = 2 - 4 \cdot 3 + 3 = -7$, which is not in $\{\pm 1, \pm 3\}$. So impossible.
- If $s_4 = 1$, then $a_6 = -M$ (allowed) and $a_5 = 2 - (M + 1) \cdot 1 + M = 1$ (allowed). Hence only $s_4 = 1$ works. This gives family 1:

$$a_6 = -M, a_5 = 1, a_4 = -M, a_3 = -1, a_2 = M, a_1 = -1, a_0 = M.$$

- **Combination (II):** $(s_0, s_1, s_2, s_3) = (1, 1, 0, 0)$. Then $a_4 = 0 - 0 + Ms_4 = Ms_4$. As before $Ms_4 = \pm M$ (the ± 1 case would force $M = 1$), so $s_4 = \pm 1$. If $s_4 = 1$, then $a_6 = -M$ and $a_5 = 0 - (M + 1) \cdot 1 + M = -1$, both allowed. If $s_4 = -1$, then $a_6 = -M - 2$, which for $M > 1$ is not in $\{\pm 1, \pm M\}$. Hence $s_4 = 1$ works. This gives family 2:

$$a_6 = -M, a_5 = -1, a_4 = M, a_3 = 1, a_2 = -M, a_1 = -1, a_0 = M.$$

- **Combination (III):** $(s_0, s_1, s_2, s_3) = (-1, -1, 0, 0)$. The same equations give $a_4 = Ms_4$, $a_6 = s_4 - (M + 1)$. The only viable choice is $s_4 = 1$, yielding $a_6 = -M$, $a_5 = -(M + 1) \cdot 1 + M = -1$, and $a_1 = 1$, $a_0 = -M$. This is family 3:

$$a_6 = -M, a_5 = -1, a_4 = M, a_3 = -1, a_2 = M, a_1 = 1, a_0 = -M.$$

- **Combination (IV):** $(s_0, s_1, s_2, s_3) = (-1, -1, -2, -2)$. Then $a_4 = -2 - (M + 1)(-2) + Ms_4 = -2 + 2M + 2 + Ms_4 = 2M + Ms_4 = M(s_4 + 2)$. Again $s_4 + 2 = \pm 1$ gives $s_4 = -1$ or $s_4 = -3$. For $s_4 = -1$: $a_6 = -1 - (M + 1) = -M - 2$; for $s_4 = -3$: $a_6 = -3 - (M + 1) = -M - 4$. Neither is in $\{\pm 1, \pm M\}$ for $M > 1$. Hence no solution.

Thus for $M > 1$ we obtain exactly three distinct families of polynomials, each valid for every odd $M > 1$.

0.0.3 Counting

- **Case $M = 1$:** 35 polynomials.
- **Case $M > 1$:** M must be odd and satisfy $1 < M < 2025^2$. Since $2025^2 = 4100625$, the largest odd below this is 4100623. The number of odd integers from 3 to 4100623 inclusive is

$$\frac{4100623 - 3}{2} + 1 = \frac{4100620}{2} + 1 = 2050310 + 1 = 2050311.$$

For each such M we have three polynomials, giving $3 \cdot 2050311 = 6150933$ polynomials.

Adding the two cases, the total number of good polynomials is

$$35 + 6150933 = 6150968.$$

6150968